

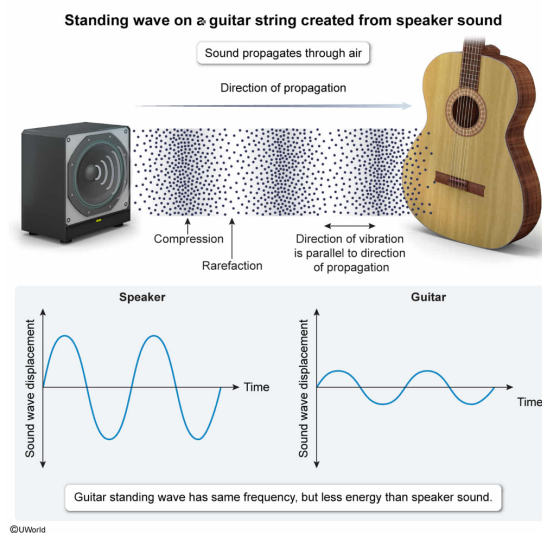
A speaker is at rest on a tabletop and produces a steady sound of constant frequency. A guitar is placed near the speaker. After some time, the speaker is turned off and a guitarist notices that one of the guitar strings is vibrating in a standing wave pattern. Which of the following statements about the standing waves on the guitar string is true?

- ☐ A. The frequency of the standing waves is lower than the frequency of the waves produced by the speaker because the sound waves lose energy before they reach the guitar string. [33%]
- ☐ B. The wavelength of the standing waves is shorter than the wavelength of the waves produced by the speaker because the sound waves lose energy before they reach the guitar string. [10%]
- ☒ C. Both waves have the same frequency because the standing waves on the string are a result of the sound waves produced by the speaker. [43%]
- ☐ D. Both waves have the same wavelength because the standing waves on the string are a result of the sound waves produced by the speaker [12%]

Correct

42% answered correctly

Explanation:



A wave is defined as perturbation that travels from one point to another. Mechanical waves (eg, sound waves, waves on a string) are perturbations created by oscillations in a **medium** such as air or a string. All waves are characterized by their **wavelength, frequency, and amplitude**.

As a wave propagates, it carries energy from one location to another. The energy contained in a wave is **directly proportional** to the square of its amplitude. When a wave moves from one medium to another, the amplitude and wavelength change but the frequency of the wave remains constant.

In this question, the speaker **creates sound** at a particular frequency. Sound is a mechanical wave formed by vibrations of air molecules. The energy in the vibrating air near the guitar creates a **standing wave** in the new medium of the guitar string with the **same frequency** as

the sound but with a different amplitude and wavelength.

(Choice A) Although the waves lose some energy as they move from the air to the guitar string, the frequency of the standing waves on the string is the same as the frequency of the sound waves.

(Choice B) The wavelength of the standing waves on the guitar string may be shorter, but this change is caused by the wave moving from one medium to another.

(Choice D) The wavelengths of the sound waves and standing waves are not the same because the air and the guitar string are different media.

Educational objective:

Mechanical waves propagate through a medium. Wavelength and amplitude depend on the type of medium, but frequency is independent of the medium.

Time spent:0 sec

Subject:
Physics

QID:403978

System:
Light and Sound

Astronauts on the Moon decide to turn on a loudspeaker outside their lunar module to broadcast Earth communications during a moonwalk. Will the astronauts hear the sound coming from the speaker while they are outside the lunar module?

- ☐ A. No, because radio waves require a medium and there is no air on the moon [15%]
- ✓ ☒ B. No, because sound is a mechanical wave requiring a medium and there is no air on the moon [74%]
- ☐ C. Yes, because sound waves, like radio waves, can travel through a vacuum [8%]
- ☐ D. Yes, if the speaker's power is turned up to a large enough value [1%]

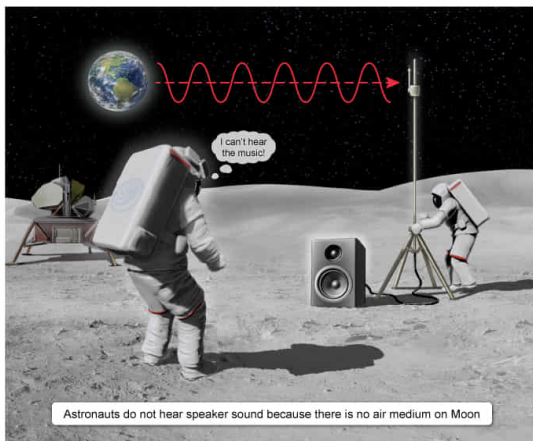
Correct

74% answered correctly

Explanation:

Mechanical waves versus electromagnetic waves

Electromagnetic waves (radio waves) propagate through vacuum from Earth to Moon



Mechanical waves (eg, sound and water waves) are disturbances that travel through a **medium** (eg, air or water). For example, **sound waves** are created by oscillating vibrations of the air that produce compressions and rarefactions (expansion) in the air as they propagate outward. In contrast, **electromagnetic waves** are perturbations in the electric and magnetic fields created by an oscillating motion of charges. Because electromagnetic fields exist independent of any medium, electromagnetic waves can propagate even when no medium is present (ie, in a **vacuum**).

In this question, the astronauts on the Moon turn on a loudspeaker to hear communications from Earth, which travel by radio (an electromagnetic wave) through the vacuum of space. The loudspeakers are designed to create sound from a vibrating source that disturbs air molecules; however, no air is present on the Moon. Therefore, the astronauts will not hear sound from the speaker because sound is a mechanical wave.

(Choice A) Radio waves are electromagnetic waves and can propagate in vacuum, but the speaker would produce *sound waves* that require a medium. The air required to carry sound waves is not present on the Moon.

(Choice C) Unlike radio waves, which can propagate in vacuum, sound waves are mechanical waves and require a medium.

(Choice D) Sound waves do not exist in a vacuum, regardless of the power of the speaker.

Educational objective:

Mechanical waves such as sound exist only in a medium, but electromagnetic waves can propagate, even in a vacuum.

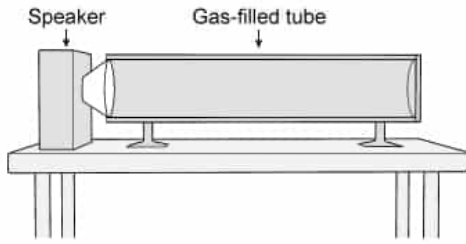
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Subject:
Physics

QID:403979

System:
Light and Sound

Researchers use a speaker to produce a sound wave with a particular frequency and amplitude in a gas-filled tube.



Which of the following changes could produce a sound wave with greater energy?

- ☐ A. Decrease the amplitude of the wave [8%]
- ✓ ☒ B. Increase the amplitude of the wave [51%]
- ☐ C. Decrease the length of the tube [34%]
- ☐ D. Increase the length of the tube [6%]

Correct

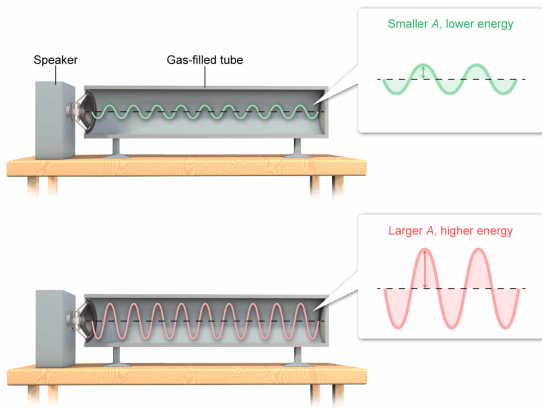
51% answered correctly

Explanation:

Sound wave energy is directly proportional to its amplitude squared

$$E \propto A^2$$

- E Energy
- A Amplitude



Mechanical waves are created by a disturbance in a medium (eg, a gas, the ocean). The disturbance is created by a force that displaces the medium at a given point. Consequently, **work** is done on the medium and as the waves propagate outward through the medium, they carry **energy**. For example, when a person flicks the end of a rope by moving one arm up and down, the person does work on the end of the rope and a wave carrying that energy travels down the length of the rope.

A larger initial displacement of the medium requires more work, and therefore produces a wave with a greater energy. As such, the energy E of a mechanical wave is **directly proportional** to the square of the **amplitude A** of the wave:

$$E \propto A^2$$

In this question, the researchers want to **produce a sound wave** with a greater energy in a particular gas medium. Because the energy of a sound wave depends on its amplitude, the

researchers must **increase the amplitude** of the wave.

(Choice A) Decreasing the amplitude of the wave will decrease the wave's energy, not increase it.

(Choices C and D) Changing the length of the tube does not affect the energy of the wave. The wave energy is determined by the speaker.

Educational objective:

The energy carried by a mechanical wave depends on its amplitude. Mechanical waves with a higher amplitude carry greater energy than waves with a lower amplitude.

Time spent:0 sec

Subject:
Physics

QID:403980

System:
Light and Sound